

- 7 The curve C has equation $y = \frac{x^2 + 2x + 1}{x - 3}$.

(a) Find the equations of the asymptotes of C .

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(b) Find the coordinates of the turning points on C .

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(c) Sketch C .

[3]

(d) Sketch the curves with equations $y = \left| \frac{x^2 + 2x + 1}{x - 3} \right|$ and $y^2 = \frac{x^2 + 2x + 1}{x - 3}$ on a single diagram, clearly identifying each curve. [4]